DATA ANALYTICS PORTFOLIO PROJECT

Maritime Logistics Data Analysis

End-to-end analytics on 15,000 container movements across 50 port terminals, 2021 to 2024: data pipeline, data model, dashboards, operational analysis and recommendations.

15,000	50	4	2021-2024
Container movements	Port terminals	Regional hubs	Period covered

BUILT WITH

Databricks · Spark SQL · Medallion architecture · Star schema · Tableau · GitHub

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Dataset: Onyx Data Challenge Maritime Dataset

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1. Project overview and objective

This portfolio project documents an end-to-end data engineering and analytics solution built on the Onyx Data Challenge Maritime Dataset. The goal is to analyze global container movement dynamics, identify port terminal bottlenecks, evaluate dwell times across key global regions, and deliver actionable operational recommendations.

Tech stack and tooling

Tool	Role in the project
Databricks (Spark SQL)	Data ingestion, cleaning, medallion architecture implementation, data enrichment and aggregation.
GitHub	Version control, documentation hosting and project portfolio repository.
Tableau	Interactive dashboard design, geospatial mapping and visual business reporting.

2. Executive summary

The dataset covers 15,000 container movements across 50 port terminals operating within four regional hubs (EMEA, APAC, AMER and LATAM) from 2021 to 2024. A Medallion Lakehouse architecture in Databricks feeds a production star schema into Tableau. The analysis highlights high-risk congestion nodes, isolates P95 dwell time anomalies, and establishes quantitative operational metrics for port efficiency.

Key findings at a glance

66.72%
of movements are classified Optimal

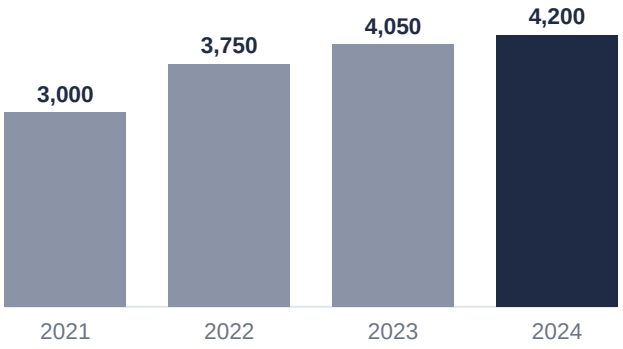
10.17%
of total duration comes from Bottleneck movements (5.83% of movements)

24.38%
of total duration comes from Underperform movements (13.89% of movements)

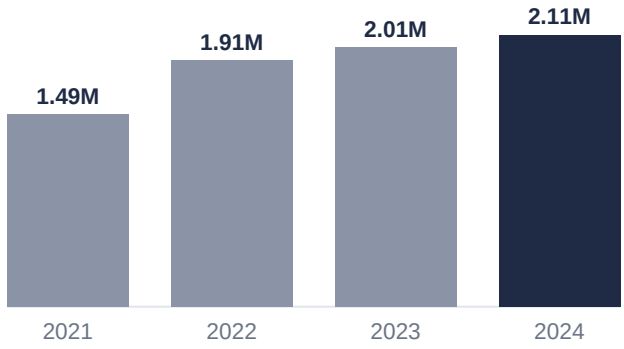
+9.5%
longer average move duration during the Suez Canal disruption (549.0h vs 501.2h)

Volume growth, 2021-2024

Total movements by year



Total containers by year



Movements and containers grew every year. Across the four years the project analyzes 15,000 movements and 7.52M containers.

Skills demonstrated

Area	What this project shows
Data engineering	Databricks, Spark SQL, Delta tables and a Bronze, Silver and Gold medallion architecture.
Data modeling	Star schema, an enriched fact table and calculated metrics based on P95 capacity and duration.
Analysis	Movement classification, bottleneck detection, year-over-year comparison and Suez Canal disruption analysis.
Visualization	Four Tableau dashboards: overview, yearly performance, terminal comparison and disruption impact.

3. Data architecture

Four operational CSV files from the Onyx Data Challenge repository (dim_terminal.csv, dim_time.csv, dim_vessel.csv and fact_cargo_movements.csv) are ingested into a Databricks environment organized in three layers and then served to Tableau.

Bronze	**Raw ingestion.** Raw landed data is stored as physical Delta tables with no transformations, preserving source integrity.
Silver	**Cleansed and enriched.** Data cleansing, schema validation and operational data enrichment across refined Delta tables.
Gold	**Business reporting.** Cleansed dimensions and fact data are consolidated into one integrated Gold view, Cargo_movement_enriched, optimized for analysis.
Tableau	**Consumption.** The Gold view is served directly to Tableau for interactive visualization, operational reporting and executive dashboards.

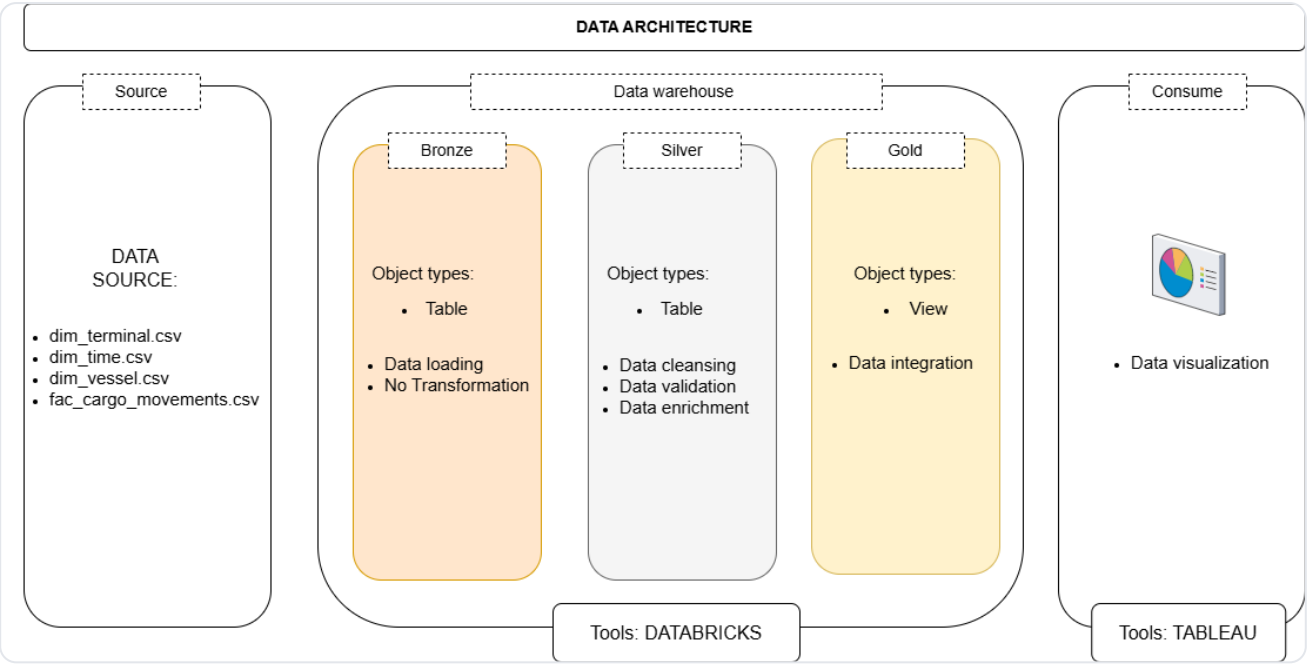
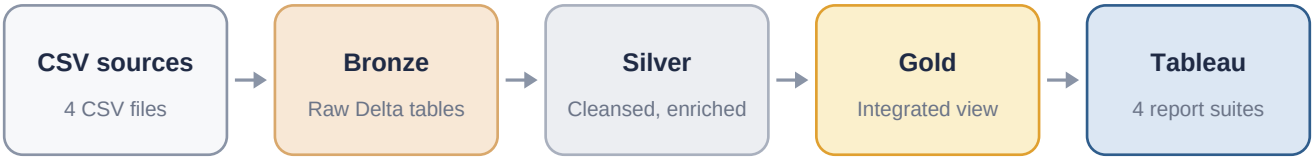


Figure 1. Data architecture: source, Databricks medallion layers and Tableau consumption.

4. Data flow

The data flow shows how source data progresses through each architectural layer into the executive dashboards.



1	**Source to Bronze.** Raw CSV files (dim_terminal, dim_time, dim_vessel, fact_cargo_movements) are loaded directly into corresponding Bronze Delta tables.
2	**Bronze to Silver.** Each entity goes through pipeline transformations (cleansing, standardizing time formats, validating coordinates) to produce production-grade Silver tables.
3	**Silver to Gold.** The cleansed dimension tables (dim_terminal, dim_time, dim_vessel) and the fact table (fact_cargo_movements) are joined and enriched into a single Gold analytical entity, Cargo_movement_enriched.
4	**Gold to reporting.** The Gold layer feeds Tableau and powers four report suites: the General Report, Yearly Report, Terminal Report and Suez Canal Report.

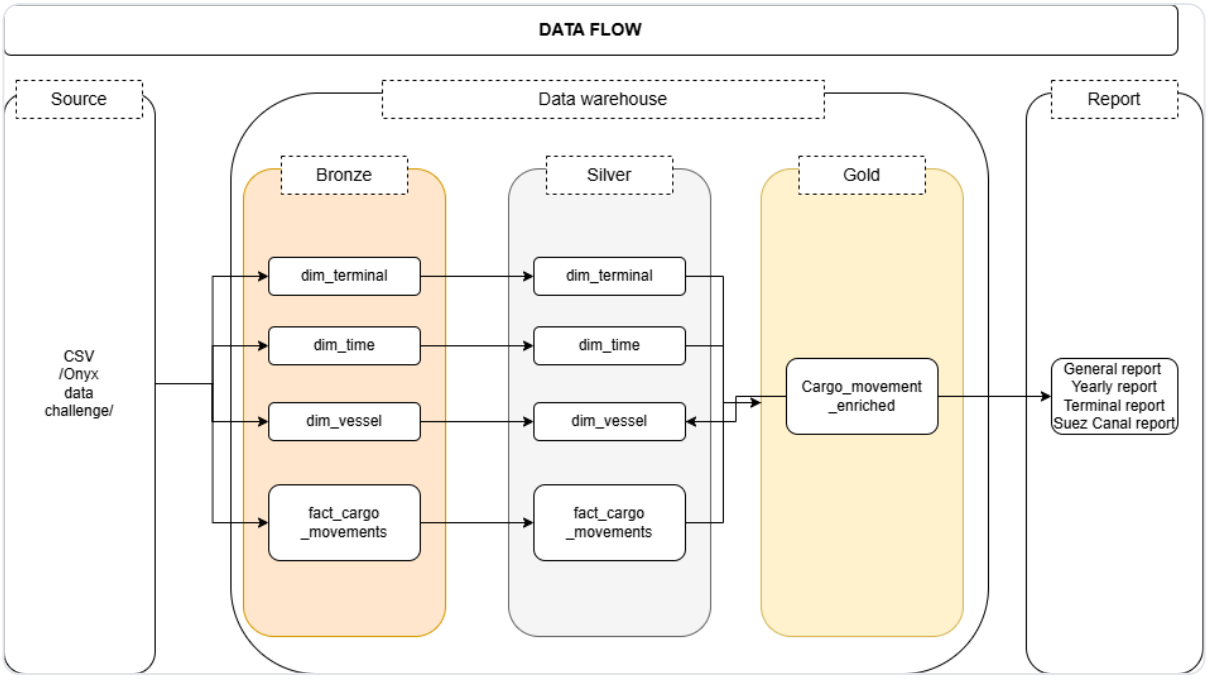


Figure 2. Data flow from CSV sources to the four Tableau reports.

5. Data model

The analytics layer uses a star schema centered on the enriched cargo movement view, linking dimensional grain with calculated operational metrics. The fact table connects to the vessel, terminal and time dimensions.

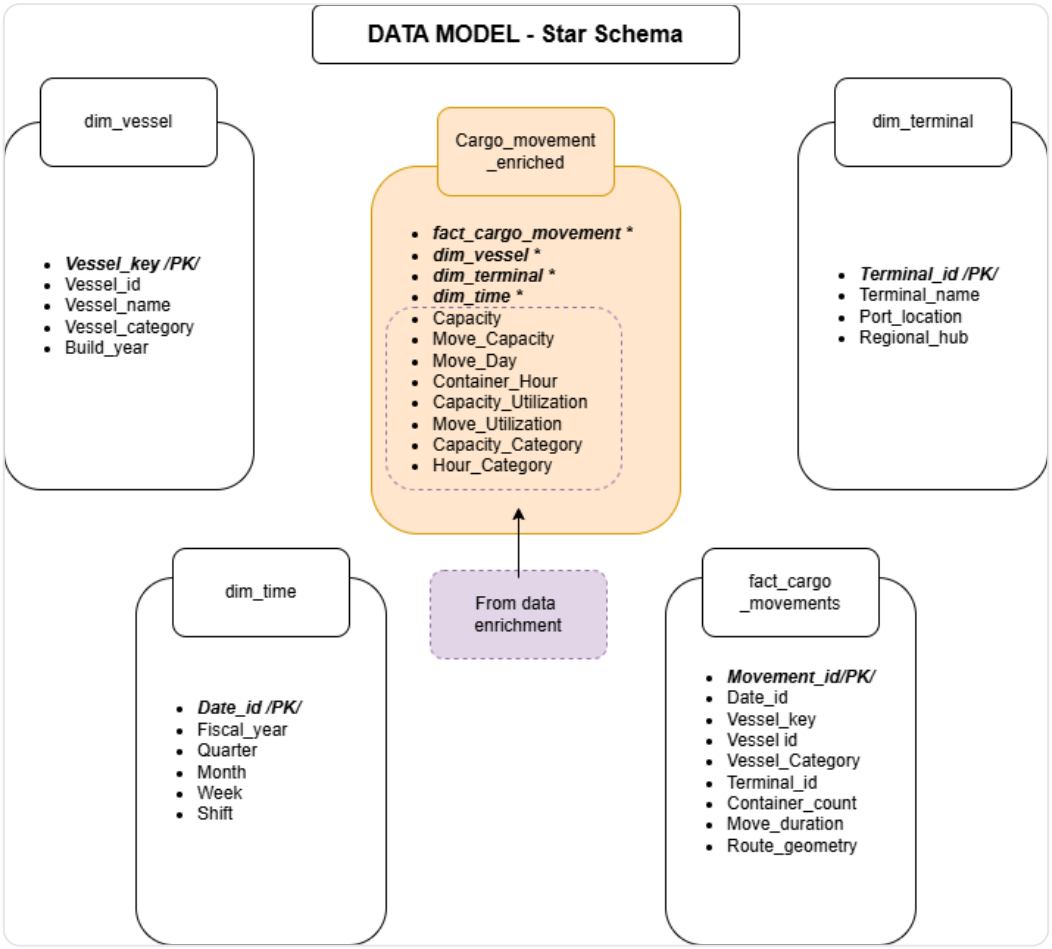


Figure 3. Star schema around Cargo_movement_enriched.

Enrichment fields in the Gold view

Capacity, Move_Capacity, Move_Day, Container_Hour, Capacity_Utilization, Move_Utilization, Capacity_Category and Hour_Category are calculated during data enrichment and added to the joined fact and dimension data. The query is shown in section 7.

6. Data dictionary

Schema overview

Table	Type	Primary key	Est. rows	Grain
dim_terminal	Dimension	terminal_id	50	One row per terminal
dim_time	Dimension	date_id	5,000	One row per date and shift combination
dim_vessel	Dimension	vessel_id	1,000	One row per vessel
fact_cargo_movements	Fact	movement_id	15,000	One row per cargo movement

Column specifications

Table	Column	Type	Description	Parameters
dim_terminal	terminal_id	integer	Terminal ID	-
	terminal_name	string	Terminal name	-
	port_location	geojson	Terminal geographic coordinates	-
	regional_hub	string	Global trade zone	EMEA, APAC, AMER, LATAM
dim_time	date_id	date	Date for the record	-
	fiscal_year	integer	Fiscal year	part = year
	quarter	integer	Fiscal quarter	part = quarter
	month	integer	Month	part = month
	week	integer	Week	part = week
	shift	string	Shift (Day/Night)	Day, Night
dim_vessel	vessel_id	integer	Vessel ID	-
	vessel_name	string	Vessel name	-
	vessel_category	string	Category of vessel	Tanker, Cargo, Passenger, Container
	build_year	integer	Year the vessel was built	uniform, 1990 to 2023
fact_cargo_movements	movement_id	integer	Unique movement identifier	-
	vessel_id	integer	Vessel identifier	-
	terminal_id	integer	Terminal identifier	-
	container_count	integer	Number of containers moved	Poisson distribution
	move_duration	float	Duration of cargo movement in hours	Normal distribution
	route_geometry	geojson	Origin and destination coordinates	-

Relationships and foreign keys

From table	From column	To table	To column	Type	Description
fact_cargo_movements	vessel_id	dim_vessel	vessel_id	Many-to-one	Links movement events to vessel attributes
fact_cargo_movements	terminal_id	dim_terminal	terminal_id	Many-to-one	Links movement events to terminal attributes

Data quality guidelines

- **Missing values.** No column in this schema permits NULL values.
- **Primary keys.** Each table has a unique, non-null primary key.
- **Temporal bounds.** All records fall between 2021-01-01 and 2024-12-31.
- **Scope.** This dictionary documents the source CSV files. In the Silver layer the fact table also carries date_id and a vessel_key surrogate key, as shown in Figure 3.

7. Related SQL queries

The enrichment query runs in the Databricks SQL environment. The classification rule is a calculated field in Tableau.

7.1 Data enrichment: building the enriched fact table

This query joins the dimension tables to the fact table and adds calculated operational metrics, creating one large enriched fact table (Cargo_movement_enriched) for Tableau.

```sql
SELECT
  f.*,
  t.terminal_name,
  t.port_location,
  t.regional_hub,
  dt.fiscal_year,
  dt.quarter,
  dt.month,
  dt.week,
  dt.shift,
  v.vessel_name,
  v.build_year,
  ct.p95 AS capacity,
  mm.move_p95 AS move_capacity,
  f.move_duration / 24 AS move_day,
  f.container_count / f.move_duration AS container_hour,
  container_count / ct.p95 AS capacity_utilization,
  CASE
    WHEN container_count / ct.p95 < 0.6 THEN 'Normal'
    WHEN container_count / ct.p95 < 0.8 THEN 'Busy'
    ELSE 'Congested'
  END AS capacity_category,
  move_duration / mm.move_p95 AS move_utilization,
  CASE
    WHEN move_duration / mm.move_p95 < 0.6 THEN 'Normal duration'
    WHEN move_duration / mm.move_p95 < 0.8 THEN 'High duration'
    ELSE 'Very high duration'
  END AS hour_category
FROM fact_cargo_movements f
LEFT JOIN terminal t
  ON f.terminal_id = t.terminal_id
LEFT JOIN dim_time dt
  ON f.date_id = dt.date_id
LEFT JOIN dim_vessel v
  ON f.vessel_key = v.vessel_key
LEFT JOIN capacity_terminal ct
  ON f.terminal_id = ct.terminal_id
LEFT JOIN move_p95 mm
  ON f.terminal_id = mm.terminal_id
```

How the query works

- **Left joins** keep every movement from the fact table, even when a matching dimension record is missing.
- **Context fields** come from the dimensions: terminal name, location and regional hub; fiscal year, quarter, month, week and shift; vessel name and build year.
- **Capacity benchmarks.** Each terminal has a P95 capacity value (capacity_terminal) and a P95 move duration (move_p95). Each movement is measured against its own terminal's benchmarks.
- **Six calculated fields** turn raw counts and durations into comparable operational metrics, as shown in the table below.

Field	Calculation	Meaning
move_day	move_duration / 24	Move duration in days.
container_hour	container_count / move_duration	Containers handled per hour of the move.
capacity_utilization	container_count / terminal P95 capacity	How much of the terminal's P95 capacity the movement used.
capacity_category	< 0.6 Normal, < 0.8 Busy, otherwise Congested	Load level of the movement.
move_utilization	move_duration / terminal P95 move duration	How long the move took compared with the terminal's P95 duration.
hour_category	< 0.6 Normal duration, < 0.8 High duration, otherwise Very high duration	Duration level of the movement.

7.2 Movement classification (Tableau calculated field)

Each movement is assigned to one of four classes by combining its capacity category and its duration category, both measured against its own terminal's P95 benchmarks. The rule is a calculated field in Tableau.

```
IF [Capacity Category] = 'Normal' AND [Hour Category] = 'Very high duration'
  THEN 'Underperformance'
ELSEIF [Capacity Category] = 'Congested' AND [Hour Category] = 'Normal duration'
  THEN 'High efficiency'
ELSEIF [Capacity Category] = 'Congested' AND [Hour Category] = 'Very high duration'
  THEN 'Bottleneck'
ELSE 'Optimal'
END
```

Label in the calculation	Name in this report	Rule	What it means
Underperformance	Underperform	Normal capacity and very high duration	A light load that still takes very long.
High efficiency	Efficient	Congested and normal duration	A heavy load handled quickly.
Bottleneck	Bottleneck	Congested and very high duration	A heavy load that takes very long.
Optimal	Optimal	All other combinations	Every movement that does not meet one of the three rules above.

The 0.6 and 0.8 cutoffs follow common port-planning rules of thumb, where yard utilization of about 60-70% is considered healthy and above about 80% signals congestion risk. They were applied to ratios against each terminal's P95 and reused for duration for consistency. These thresholds are an analytical assumption, and results may change with different cutoffs.

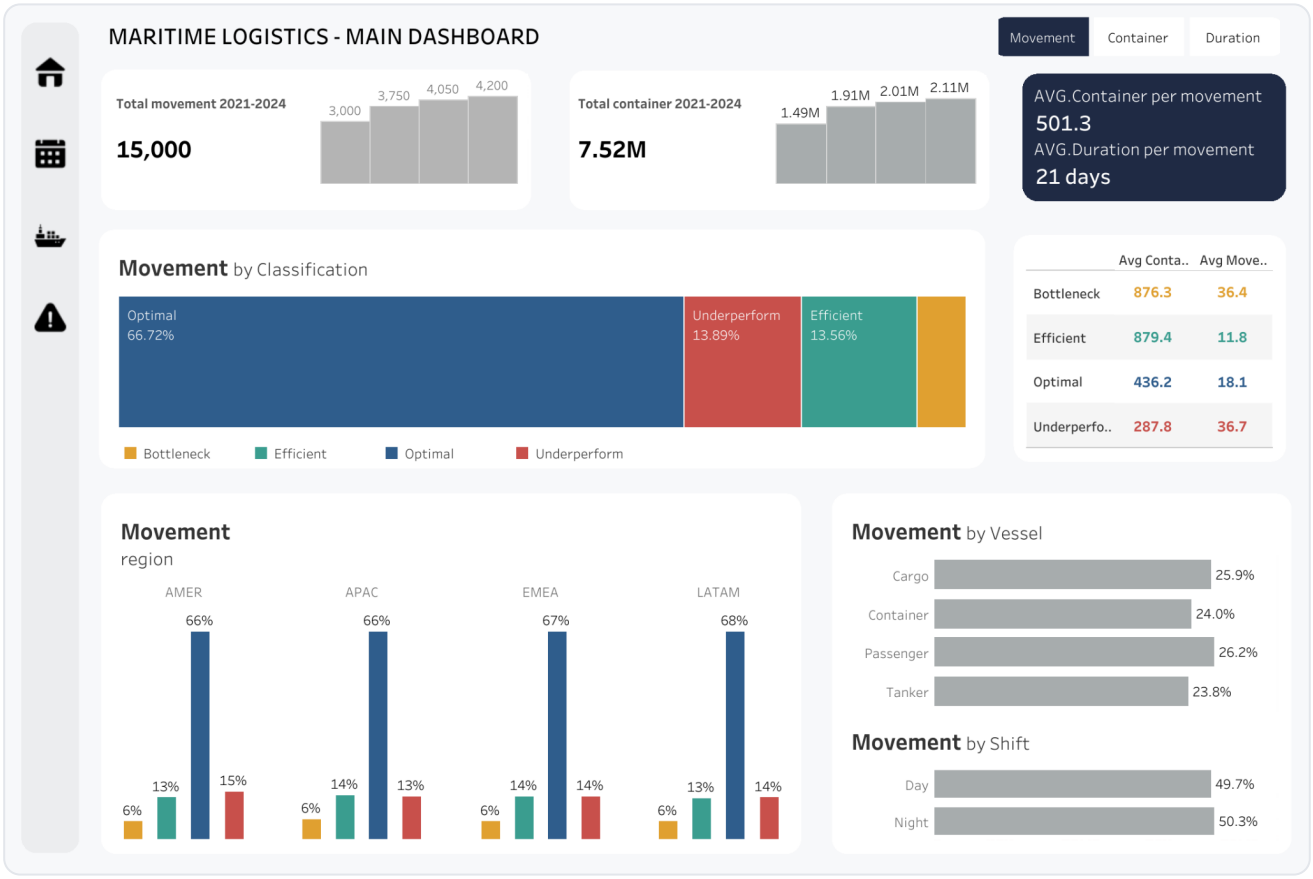
Sources for the rules of thumb: Philippine Ports Authority guidance on yard utilization (70% optimal, above 80% at risk of congestion), as reported by Malaya Business Insight; Park et al., Simulation and Analysis of Container Terminals Operations: Case Study of Korean Ports (operator survey, about 60% yard occupancy).

8. Dashboards

Four Tableau dashboards serve the Gold view: a main overview, a yearly performance view, a terminal comparison and a Suez Canal disruption analysis.

8.1 Main dashboard

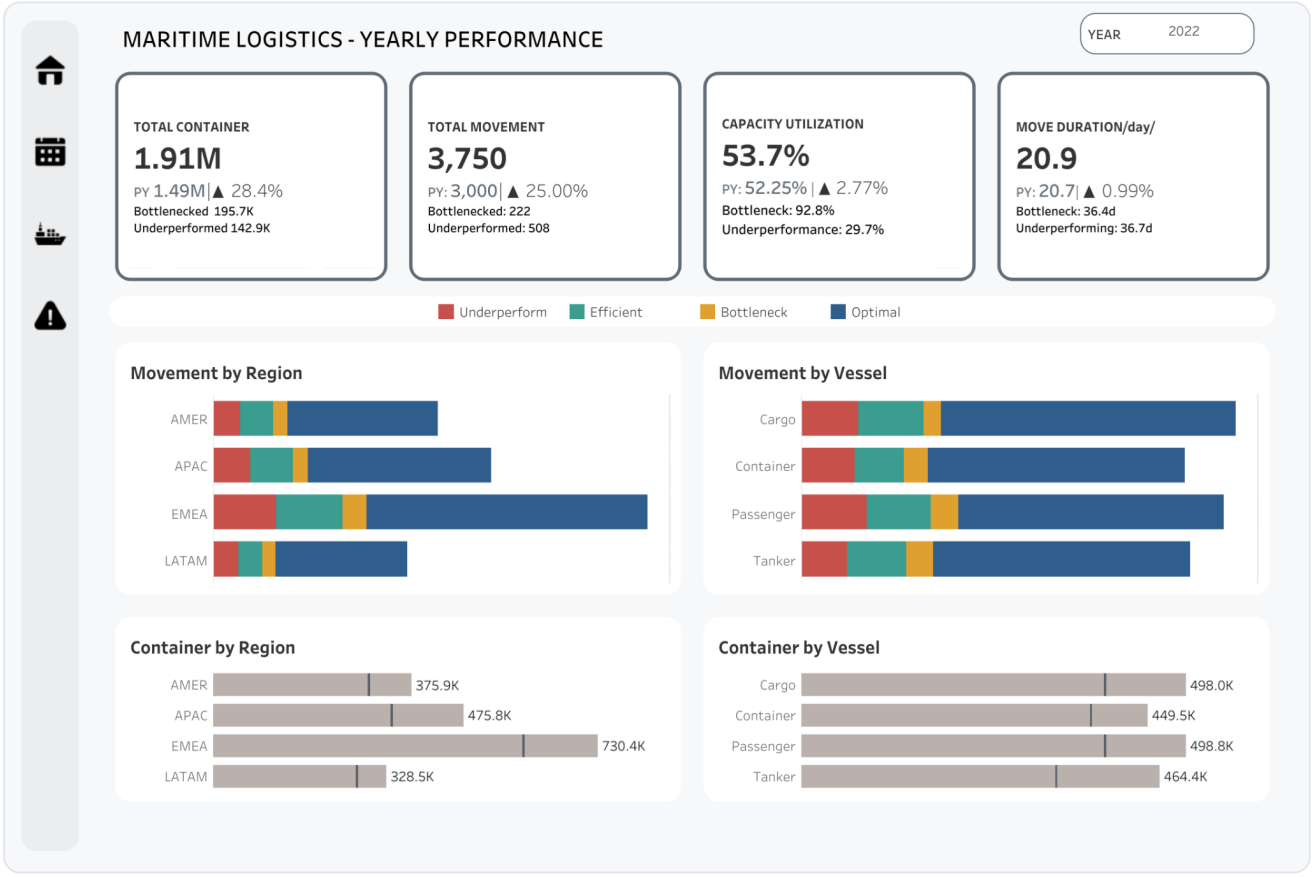
Overall view of movement, container and duration performance across 2021-2024.



- 15,000 movements and 7.52M containers were handled in total. Both grew every year: movements from 3,000 in 2021 to 4,200 in 2024, and containers from 1.49M to 2.11M.
- 66.72% of movements are Optimal, followed by Underperform (13.89%), Efficient (13.56%) and Bottleneck (5.83%).
- Bottleneck and Underperform movements take 36.4 and 36.7 days on average, against 11.8 days for Efficient and 18.1 days for Optimal movements.
- Each region shows a similar mix (66% to 68% Optimal). Movements are spread almost evenly across vessel types (23.8% to 26.2%) and between day (49.7%) and night (50.3%) shifts.

8.2 Yearly performance dashboard

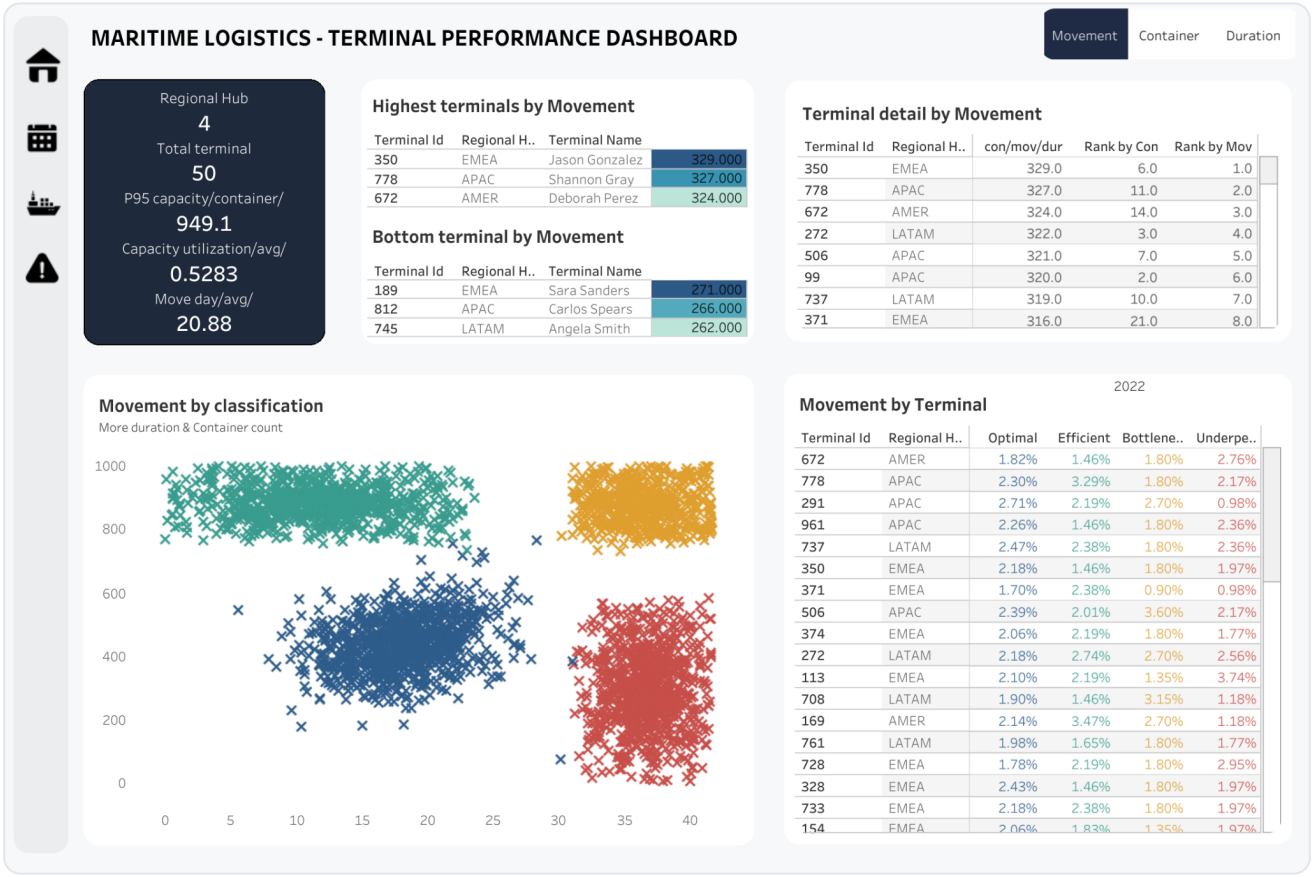
Year-over-year performance for the selected year (the 2022 view is shown), with prior-year comparisons on every KPI.



- Total containers reached 1.91M (+28.4% against 1.49M in 2021) and total movements 3,750 (+25.0% against 3,000).
- Capacity utilization was 53.7% (+2.77% against the prior year) and average move duration 20.9 days (+0.99%).
- Each KPI card also shows the Bottleneck and Underperform volume behind it, for example 222 Bottleneck and 508 Underperform movements.
- Movement is broken down by region and by vessel type as stacked classification bars, with container volume by region and vessel type below. EMEA is the largest region at 730.4K containers.

8.3 Terminal performance dashboard

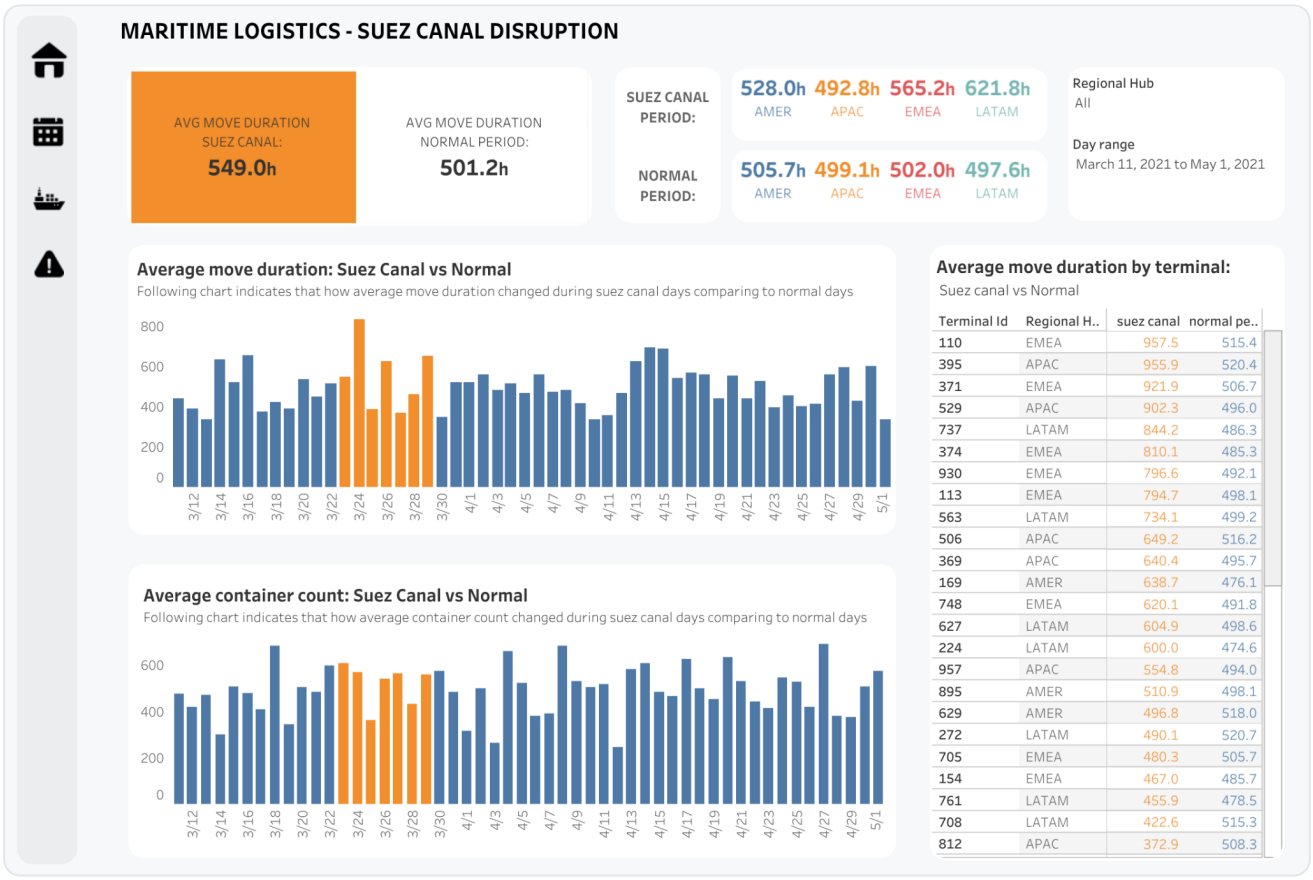
Comparison of the 50 terminals across the four regional hubs.



- The summary panel shows 4 regional hubs, 50 terminals, a P95 capacity per container of 949.1, average capacity utilization of 0.5283 and an average move duration of 20.88 days.
- Terminals with the most movements are 350 (EMEA, 329), 778 (APAC, 327) and 672 (AMER, 324). Those with the fewest are 189 (EMEA, 271), 812 (APAC, 266) and 745 (LATAM, 262).
- The scatter plot of duration against container count separates the classes: Efficient movements combine high container counts with short durations, Bottleneck movements combine high counts with long durations, and Underperform movements combine long durations with low counts.
- A detail table gives each terminal's classification mix by year (2022 shown) and its rank by container count and by movement.

8.4 Suez Canal disruption dashboard

Comparison of move duration and container count during the Suez Canal disruption against normal days, for March 11 to May 1, 2021.



- Average move duration rose from 501.2h in the normal period to 549.0h during the disruption.
- EMEA (565.2h against 502.0h) and LATAM (621.8h against 497.6h) were the most affected. AMER was slightly higher (528.0h against 505.7h) and APAC was not affected (492.8h against 499.1h).
- The daily charts highlight the disruption days in orange for both average move duration and average container count.
- The terminal table lists average duration in both periods. The longest during the disruption were terminals 110 (EMEA), 395 (APAC) and 371 (EMEA), each above 920h and about 1.8 times their normal level.

9. Analytical results

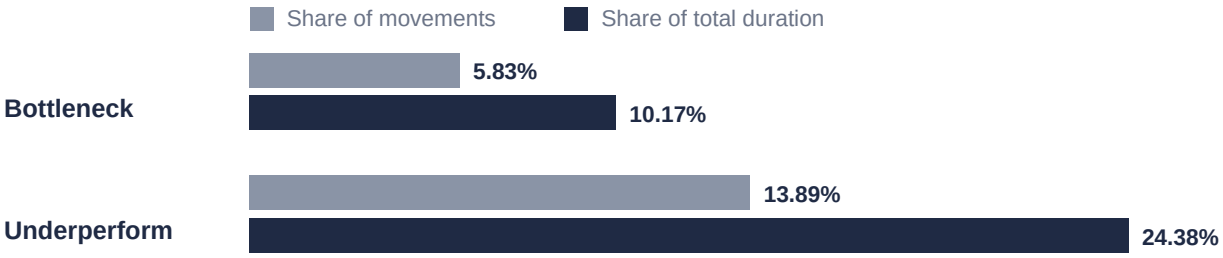
Movements were classified by duration and container volume to evaluate operational performance.

Performance by classification

Classification	Share of movements	Avg containers per movement	Avg move duration (days)	Share of total duration
Optimal	66.72%	436.2	18.1	-
Efficient	13.56%	879.4	11.8	-
Bottleneck	5.83%	876.3	36.4	**10.17%**
Underperform	13.89%	287.8	36.7	**24.38%**

Optimal and Efficient movements together account for the remaining 65.45% of total duration.

Where the duration concentrates



Bottlenecked movements account for 5.83% of total movements but 10.17% of total duration. Underperformed movements represent 13.89% of movements but 24.38% of total duration, while handling relatively lower container volumes.

This pattern points to inefficiency in Underperform movements, where longer processing times are not matched by higher throughput. Because the dataset contains limited operational variables, the exact root causes cannot be fully determined.

Suez Canal disruption impact

The Suez Canal disruption period had a significant effect on operational performance. Average movement duration increased from 501.2 hours to 549.0 hours, with EMEA and LATAM the most affected regions.

Region	Normal period (h)	Suez period (h)	Change (h)	Change (%)
AMER	505.7	528.0	**+22.3**	**+4.4%**
APAC	499.1	492.8	**-6.3**	**-1.3%**
EMEA	502.0	565.2	**+63.2**	**+12.6%**
LATAM	497.6	621.8	**+124.2**	**+25.0%**
All regions	**501.2**	**549.0**	**+47.8**	**+9.5%**

10. Recommendation

> **Prioritize reducing delays in Underperform movements.** They account for a disproportionate share of total duration (24.38% from 13.89% of movements) and should be investigated further. Targeted optimization in this area could potentially contribute to up to a 15% reduction in overall movement duration.

Limitations. The dataset contains limited operational variables, so this analysis shows where delays concentrate rather than why they occur. Explaining the causes would require additional operational data.